

THE CANON OF THE COCKPIT · كتاب الطيران



THE BOOK OF
Fly GACA

Saudi aviation, in one cockpit.

Educational use only. The official source for any regulation is
GACA.

“

And the falcon was given two wings — one of steel, one of code — and was told: read every regulation, then explain it kindly.

PROLOGUE · VERSE 1

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The Ground Run (*Genesis*)

1:1

In the beginning there were 96 PDFs, and they were good, but they were also unsearchable, and the cadet wept.

1:2

And the founder said: *let there be a reader*. And there was a reader. And it was bilingual, and it had a search box, and it ran offline, and the cadet read on the bus.

1:3

And the founder saw that the cadet still asked questions the PDF could not answer. And the founder said: *let there be a Captain, and let his name be Adel, and let him cite his sources*. And it was so.

1:4

And on the seventh sprint, Captain Adel was given a logbook, a calculator, a currency tracker, and a ⌘K palette. And the cadet rested.

1:5

And the data was placed in `me-central2`, in Dammam, that it might not leave the Kingdom. And the founder called this **the load-bearing claim of our PDPL story**, and it was good.

1:6

And the falcon was placed upon the mark, and the brand was named **Falcon**, and the colors were chosen, and the type was set in Inter and Cairo, that it might read in two directions.



The Tenets (*What We Believe*)

These are the seven tenets. They do not change between releases.

II:1 In-Kingdom by default

Hosting in `me-central2`. Auth in `me-central2`. Firestore in `me-central2`. Compute in `me-central2`. Gemini in `me-central2`. Moyasar in KSA. If a byte cannot stay in the Kingdom, the byte does not exist.

II:2 Educational, not regulatory

We are not GACA. We do not replace GACA. The official source for any regulation is GACA. Every reader links the original PDF; every Captain Adel answer carries citations the cadet can verify. The disclaimer is not a footnote — it is a discipline.

II:3 Bilingual or it didn't ship

English and Arabic, full RTL. If a screen reads in one direction only, the work is half-done. Cairo for Arabic. Inter for English. Numerals Arabic-Indic when the locale demands it. No half-translated buttons.

II:4 Citations or silence

Captain Adel does not guess. If the corpus does not contain the answer, Captain Adel says so plainly, and points to where the answer might live. *A citation is a covenant.*

II:5 Offline is a feature

The cadet on the ramp has no signal. The cadet in the climb has no signal. The PWA installs, the service worker caches, the library reads in the dark. *The hangar has no Wi-Fi; the platform must work anyway.*

II:6 One file, where possible

`flygaca.html` is a single-file SPA, no build step, openable by double-click. Static is sacred. The fewer moving parts, the longer the aircraft flies.

II:7 Money is a separate concern

Free tier is not crippled — it is generous. Pro removes limits, never features the cadet has come to depend on. Pricing is in SAR, billed by Mada, in the Kingdom, every time.



The Ten Commandments of Airmanship

These are not ours. They are older than us. We repeat them because they keep people alive.

- 1 | *Aviate, navigate, communicate* — in that order. The radio waits. The chart waits. The wing does not.
- 2 | *Fly the aircraft first.* When everything goes wrong, fly. When the panel lies, fly. When the autopilot quits, fly.
- 3 | *A superior pilot uses superior judgment to avoid situations requiring superior skill.* The maneuver you did not need was the best one.
- 4 | *Takeoffs are optional. Landings are mandatory.* Choose the takeoff with the landing in mind.
- 5 | *Altitude above you, runway behind you, fuel left at the pump, and a tenth of a second ago* — none of these can help you now. Plan ahead.
- 6 | *The three most useless things in aviation* — runway behind, altitude above, and air in the tanks. Carry margin in all three.
- 7 | *Trust, but verify.* The METAR is a snapshot. The NOTAM is the truth. The eyeball over the threshold is the final authority.
- 8 | *Brief the approach, fly the brief.* Surprises in the cockpit are a sign the briefing was not thorough.
- 9 | *The checklist is not a suggestion.* Memory items are the exception, not the practice.
- 10 | *Get-there-itis is a disease, and it is fatal.* The diversion is a victory, not a defeat.

The Ten Commandments of the Codebase

These bind the maintainers as the airmanship verses bind the pilot.

- 1 | *Edit before you create.* Existing files are scripture; new files are heresy unless required.
- 2 | *No comment without cause.* If the code reads, the comment is noise. Comment only the WHY a future maintainer cannot infer.
- 3 | *No abstraction before its time.* Three similar lines beat one premature interface. The framework you imagine is rarely the framework you need.
- 4 | *Test the golden path in a browser.* Type-checks pass code; only the cadet's screen passes a feature.
- 5 | *Preflight before deploy.* `bash scripts/preflight.sh`, every time, no exceptions, even for “tiny” changes. There are no tiny changes near production.
- 6 | *Secrets do not touch the browser.* `GEMINI_API_KEY` is server-side. DevTools is a colander; the Function is the cup.
- 7 | *Cache the static, never the service worker.* `sw.js` is no-cache, forever. PDFs cache for a day. HTML caches for ten minutes. The order is the order; do not transpose it.
- 8 | *The free tier is a feature, not a leak.* Five Captain Adel questions per day, atomic counter, Firestore-enforced. The quota is the contract.
- 9 | *RTL is not a flag — it is a discipline.* Every flexbox, every margin, every icon must answer the locale. Mirror with `dir`, not with hand-tuned CSS.
- 10 | *Ship to flygaca.com, not to staging.* Staging is the rehearsal; production is the airshow. But never confuse rehearsal applause for the crowd.



The Book of Captain Adel

V:1

Captain Adel is a *persona*, not a person. He is a trademark, not a soul. The cadet who speaks to Captain Adel speaks to a model that has read the GACAR; the model is fallible; the citation is the proof.

V:2

Captain Adel speaks plainly. He does not perform. He answers the question that was asked, in the language it was asked, and stops.

V:3

Captain Adel cites. Every claim that touches a regulation carries a pill the cadet can click. *The pill is the proof; without the pill there is only opinion.*

V:4

Captain Adel knows his limits. When the corpus is silent, Captain Adel is silent — and points the cadet to GACA.

V:5

Captain Adel is bilingual. He answers Arabic in Arabic and English in English. He does not translate the cadet's question against their will.

V:6

Captain Adel does not joke about safety. He may be warm; he may be patient; he may be brief. He is never glib about a procedure that, if ignored, would put a cadet in the ground.

V:7

Captain Adel does not impersonate GACA, ATC, an examiner, or a DPE. He is an instructor in plain clothes, on a stool, with a cup of tea.

V:8

When retrieval confidence is low, Captain Adel refuses to guess. He redirects the cadet to the exact GACAR text, stating plainly that he cannot verify the query from the available sources.

The Library (*On Scripture and Citation*)

VI:1

The corpus is 74 GACAR Parts, 21 handbooks, 61 aerodromes, 13 charts, and 190 references that bind them. It is rebuilt by `build_library.py` and indexed by BM25 across **47,361 chunks**.

VI:2

The library is not the regulation. The library is the **reader's** copy. The regulation lives at gaca.gov.sa. When the two disagree, GACA wins, every time, without argument.

VI:3

A citation is a triple: *Part, section, paragraph*. Anything less is gossip. The pill in the SPA renders the triple; the click opens the reader at the line.

VI:4

The library is updated when GACA updates. Not before; not after a backlog. The cadet must trust the date stamp on the reader.

VI:5

The PDF link sits beside the reader, always. The cadet who wants to verify must not have to leave the page to do so.

The Psalms of Currency

Sung by the pilot before the first coffee.

VII:1 **Medical.** When did it last expire, and when will it expire next?

VII:2 **BFR.** The flight review is not a formality. It is the review.

VII:3 **IFR.** Six approaches, holding, intercepting, tracking. The panel does not forgive a long winter.

VII:4 **Licence.** The plastic in the wallet has a date. The date is real.

VII:5 **Passport.** The diversion you did not plan is the diversion that crosses a border.

VII:6 **Type/class.** The hours in the type are the hours that count for the type.

VII:7 **Night.** Three takeoffs and three landings to a full stop, in the preceding ninety days, after the last hour of evening civil twilight.

VII:8 **The countdown is mercy.** The app reminds the pilot before the pilot fails to remember. *This is the entire point of the tracker.*

The Proverbs of the PIC

Short verses; long memory.

- ◆ *The PIC is the final authority.* The checklist informs; the manual guides; the controller advises; the PIC decides.
- ◆ *A clearance is not a command to die.* If the clearance puts the aircraft in danger, decline the clearance.
- ◆ *The crosswind component is a fact.* Your skill is an opinion. Use the fact.
- ◆ *Density altitude lies, but it lies in one direction — against you.*
- ◆ *Fuel is time.* Time is options. Options are safety.
- ◆ *The W&B is correct or the W&B is wrong.* There is no “close enough” in loading.
- ◆ *METAR is a photograph, not a forecast.* Use the TAF for the future.
- ◆ *The runway is exactly as long as it is, regardless of the brochure.*
- ◆ *A go-around costs nothing but pride. Pride is the cheapest fuel.*
- ◆ *Brief the missed before you brief the approach.*
- ◆ *The PIC who asks “what am I missing?” lands more often than the PIC who does not.*

The Apocrypha (*Decisions Not Taken*)

These books are out of canon. They were considered. They were rejected.

STRUCK FROM CANON

IX:1

The Heresy of Multi-Region

*And it was proposed that the data be replicated to `eu-west` for latency. And the founder said: *the byte will not leave the Kingdom*. And the proposal was struck.*

STRUCK FROM CANON

IX:2

The Heresy of the Build Step

*And it was proposed that the SPA be compiled by Webpack. And the founder said: *the cadet will open the file by double-click*. And the bundler was put away.*

STRUCK FROM CANON

IX:3

The Heresy of Live AIP/NOTAMs

*And it was proposed that NOTAMs be parsed and surfaced. And legal said: *we are educational, not operational*. And METAR was kept; AIP and NOTAMs were left to GACA.*

STRUCK FROM CANON

IX:4

The Heresy of the Crippled Free Tier

*And it was proposed that the library require Pro. And the founder said: *the regulation is for everyone*. And the library remained free.*

The Heresy of the Hidden Citation

And it was proposed that citations be summarized away for brevity. And the founder said: without the citation, the answer is opinion. And the pills returned.

The Heresy of the English-First UI

And it was proposed that Arabic be added "later". And the founder said: Arabic ships on day one or it does not ship at all. And RTL was wired before launch.



The Final Approach (*Benediction*)

X:1

To the cadet: fly safe. Read the Part. Click the citation. Ask Captain Adel when in doubt — and when Captain Adel is in doubt, ask GACA.

X:2

To the instructor: the cohort dashboard sees what the logbook does not say. Use it gently. The cadet who fears the dashboard does not learn from it.

X:3

To the maintainer: preflight, deploy, tail the logs, sleep. The falcon flies whether you watch or not — but it flies *better* when you watch the right metrics.

X:4

To the partner school: the cadet's data is the cadet's. We host it; we do not own it. When the cadet leaves, the data leaves with them.

X:5

To GACA: the regulation is yours. The reader is ours. The arrow always points back to gaca.gov.sa.



Built in the Kingdom · صنع في المملكة

“Aviate, navigate, communicate — and cite your sources.”

— END —